Lecture 1: Introduction & Vector Space Model

Information Retrieval — Lecture Notes

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Information Retrieval Course

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Abstract

These lecture notes accompany Lecture 1 of the Information Retrieval course. We introduce the fundamental concepts of Information Retrieval (IR), distinguishing it from database systems and natural language processing. We explore the core challenges of scale and semantic understanding before diving into the two classical retrieval models: the **Boolean Retrieval Model** and the **Vector Space Model (VSM)**. Key topics include TF-IDF term weighting and cosine similarity.

Contents

1 What is Information Retrieval?

Definition 1.1 (Information Retrieval)**.** Information Retrieval (IR) is the field concerned with finding material (usually documents) of an unstructured nature (usually text) that satisfies an information need within large collections.

In essence, IR is about bridging the gap between a user's *information need* and a massive collection of items. Unlike structured data retrieval in databases, IR deals with the inherent ambiguity of natural language.

1.1 IR vs. Database Systems

The primary difference between IR and Database Management Systems (DBMS) lies in the structure of the data and the precision of the query.

Aspect	Database Systems	IR Systems
Data	Structured (tables, schemas)	Unstructured (text, documents)
Queries	Precise (SQL)	Vague (natural language)
Matching	Exact	Approximate/Ranked
Result	All matching records	Top-k relevant documents
Failure	No results or all results	Ranked list (best effort)

Key Takeaway

Databases ask: *"Give me all records matching X"*. IR asks: *"Give me the most relevant documents about X"*.

1.2 IR vs. Natural Language Processing (NLP)

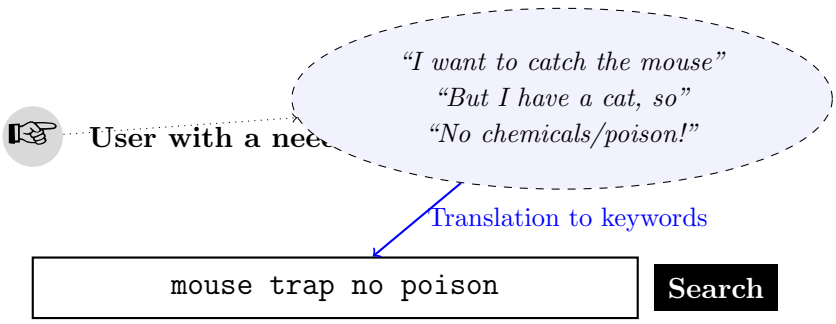
While IR and NLP share tools (like tokenizers and embeddings), they have different focal points:

- **NLP (Understanding):** Focuses on language representation, syntax, semantics, and generation.
- **IR (Retrieval):** Focuses on information seeking, scale, and ranking.

2 The Core Challenges of IR

2.1 The Semantic Gap

The most significant challenge in IR is the *semantic gap*—the difference between the keywords used in a query and the underlying intent of the user.



For a query like `mouse trap no poison`, a simple keyword matching system might return documents discussing how poison traps work because the words "poison" and "trap" appear. However, the user's intent is to find *mechanical* or *humane* alternatives.

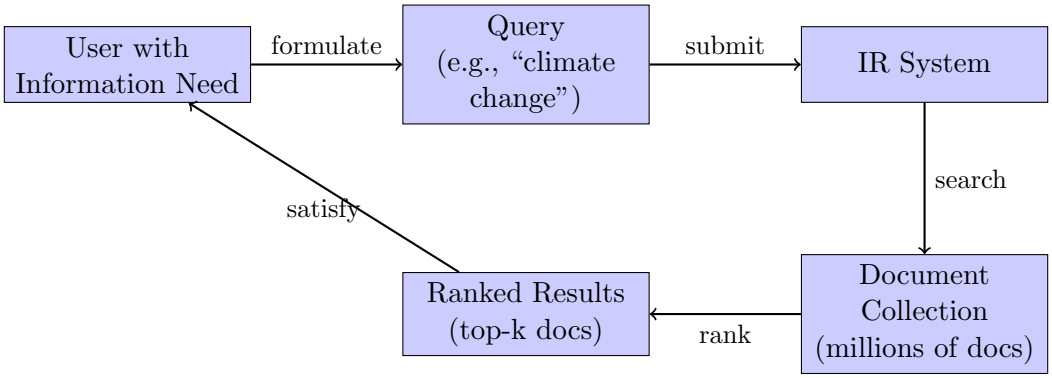
2.2 Scale and Freshness

Modern IR systems must operate at a scale that is difficult to comprehend:

- **Scale:** Google indexes over 100 trillion pages.
- **Throughput:** Billions of queries per day.
- **Latency:** Sub-second response times are mandatory.
- **Freshness:** The index must reflect the state of the web in near real-time.

3 The Information Retrieval Problem

The retrieval process follows a cycle where a user formulates a need into a query, the system searches a collection, and the user evaluates the ranked results.



This process raises three fundamental questions that define the design of any IR system:

1. How do we represent documents and queries?
2. How do we match queries to documents efficiently?
3. How do we rank documents by relevance?

4 The Boolean Retrieval Model

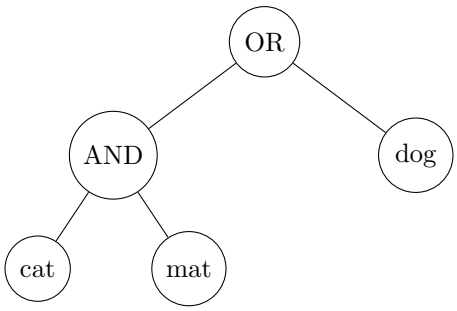
The Boolean model is the simplest retrieval model. It treats documents as sets of terms and queries as Boolean expressions.

Definition 4.1 (Boolean Retrieval)**.** In the *Boolean retrieval model*, a document is retrieved if it satisfies a Boolean expression of query terms using operators `AND`, `OR`, and `NOT`.

4.1 Query Processing with Set Operations

Queries are processed as set intersections, unions, and differences of posting lists (the sets of document IDs containing each term).

Example 4.1 (Boolean Query Tree)**.** Consider the query: `(cat AND mat) OR dog`. The system first intersects the sets of documents containing "cat" and "mat", then unions the result with the set of documents containing "dog".



4.2 The Feast or Famine Problem

Boolean retrieval suffers from a binary view of relevance: a document is either in or out.

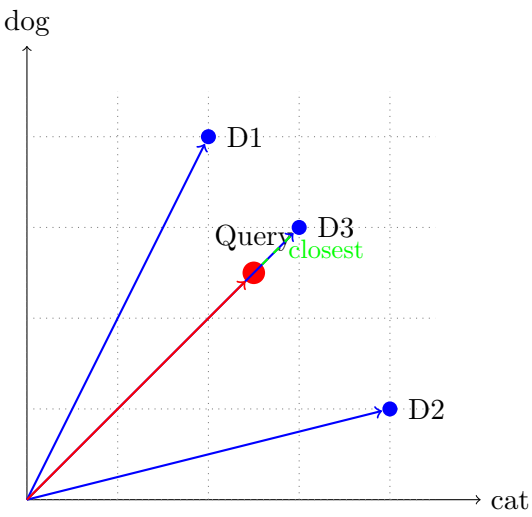
- **Famine:** A query with too many `AND` terms may return 0 results if even one minor term is missing (too restrictive).
- **Feast:** A broad query may return millions of results with no way to identify the best ones (too permissive).

Key Takeaway

Boolean retrieval lacks **ranking** and **partial matching**. This motivates the transition to weighted models like the Vector Space Model.

5 The Vector Space Model (VSM)

The Vector Space Model represents documents and queries as vectors in a high-dimensional space, where each dimension corresponds to a term in the vocabulary.



5.1 Term Weighting: TF-IDF

To create meaningful vectors, we must assign weights to terms. The standard approach is **TF-IDF** (Term Frequency-Inverse Document Frequency).

Definition 5.1 (TF-IDF)**.** The weight of term t in document d is:

$$w_{t,d} = \mathrm{tf}(t,d) \times \log\left(\frac{N}{\mathrm{df}(t)}\right)$$

where:

- $\mathrm{tf}(t, d)$ is the number of occurrences of term t in document d.
- N is the total number of documents in the collection.
- $\mathrm{df}(t)$ is the number of documents containing term t.

Intuition:

- **TF** rewards terms that appear often in a document (local importance).
- **IDF** penalizes terms that appear often in the collection (stop words like "the") and rewards rare, discriminative terms (global importance).

5.2 Worked Example: TF-IDF Calculation

Example 5.1 (TF-IDF Weights)**.** Collection of 10,000 documents. Document d contains: "neural" (5 times), "network" (3 times), and "the" (50 times).

Term	tf(t,d)	df(t)	idf(t)	tf-idf
neural	5	100	$\log(10000/100) = 2.0$	10.0
network	3	500	$\log(10000/500) = 1.3$	3.9
the	50	9000	$\log(10000/9000) = 0.05$	2.5

Despite appearing 10 times more often, "the" receives a lower weight than "neural" because it carries less discriminative information.

6 Cosine Similarity

In VSM, the relevance of a document to a query is measured by the *cosine* of the angle between their vectors.

Definition 6.1 (Cosine Similarity)**.** The similarity between query vector $\vec{q}$ and document vector $\vec{d}$ is:

$$\mathrm{sim}(\vec{q}, \vec{d}) = \frac{\vec{q} \cdot \vec{d}}{\|\vec{q}\| \|\vec{d}\|} = \frac{\sum_t q_t \times d_t}{\sqrt{\sum_t q_t^2} \times \sqrt{\sum_t d_t^2}}$$

Key Takeaway

Cosine similarity normalizes for document length. Without this normalization, long documents would naturally have larger dot products simply because they contain more words, regardless of their actual relevance.

7 VSM vs. Boolean Retrieval

Aspect	Boolean	Vector Space
Matching	Exact (in/out)	Partial (weighted)
Ranking	No	Yes (by similarity)
Robustness	Brittle	Graceful degradation
Term weighting	Binary	TF-IDF
User friendliness	Expert users	General users

Remark 7.1 (Bag of Words Assumption)**.** Both Boolean and simple VSM models share the **bag-of-words** assumption: they treat documents as unordered collections of words, ignoring grammar and word order. This limitation is addressed in later lectures using phrase matching and transformer-based models.